Week 8

1) A key limitation of MapReduce that motivated Spark was:
A. Lack of fault tolerance
B. High cost of repeatedly writing intermediate data to disk
C. Inability to run on commodity clusters
D. Absence of parallelism

2) Spark was originally developed at:
A. Google Research
B. Microsoft Research
C. UC Berkeley AMPLab
D. Apache Foundation

3) An RDD in Spark is best described as:
A. An immutable, partitioned collection of records
B. A mutable table stored in HDFS
C. A centralized in-memory cache
D. A disk-only dataset

4) RDD fault tolerance is primarily achieved using:
A. Replication of all data
B. Write-ahead logging
C. Lineage information
D. Distributed locking
Answer: C

5) Which of the following is an example of an *action* in Spark?
A. map()
B. filter()
C. join()
D. count()

6) In PageRank, each page contributes:
A. Its full rank to every neighbor
B. rank × |neighbors| to each neighbor
C. rank / |neighbors| to each neighbor
D. A constant value to neighbors

7) Which Spark library is used for graph processing?
A. Spark SQL
B. Spark Streaming
C. MLlib

D. GraphX

8) The Gather-Apply-Scatter (GAS) model in GraphX is used to:
A. Execute SQL queries
B. Perform iterative graph computations
C. Replicate graph data
D. Encrypt graph edges

9) Apache Kafka is best described as:
A. A relational database
B. A real-time distributed messaging system
C. A batch analytics engine
D. A file system

10) In Kafka's data model, messages are grouped into:
A. Streams
B. Queues
C. Topics
D. Tables

11) In Kafka, the server process that stores and serves messages is called a:
A. Producer
B. Consumer
C. Broker
D. Connector

12) Kafka supports which messaging models?
A. Only queue-based
B. Only publish-subscribe
C. Both queue and publish-subscribe
D. Request-reply only

13) A Kafka topic with replication factor N can tolerate:
A. N failures
B. N−1 broker failures
C. Only one failure
D. No failures